

Weight control in the management of hypertension

World Hypertension League¹

This article, which includes a brief description of the mechanisms and some epidemiological findings in obesity and high blood pressure, sums up present knowledge on a complex subject and provides guidance to medical practitioners on the management of obese hypertensive patients. Weight reduction, together with drug therapy in severe and moderate hypertension, and other non-pharmacological methods and continuing observation in mild hypertension are the essential measures to be applied. In addition to the lowering of blood pressure, weight loss offers several other metabolic and haemodynamic benefits.

Like the previous paper on self-measurement of blood pressure (1), this text has been endorsed by representatives of the member organizations of the World Hypertension League, thus presenting wide international consensus. In order to reach such consensus, some simplifications were unavoidable.

Obesity and hypertension

Mechanisms

The increased metabolic demand of obese individuals is known to raise the cardiac output (expressed in absolute terms) although the cardiac index, which is a relative measure, may be normal or even decreased. The increased cardiac output is usually due to an increase in stroke volume; the plasma and total blood volumes are also usually raised. When vascular resistance increases (as it does in the pathogenesis of essential (primary) hypertension), a high cardiac output related to obesity may act as an important factor contributing to the development of

hypertension. Other mechanisms that may be involved are: genetic and environmental determinants of body weight and blood pressure; cell membrane abnormalities; and some endocrine and metabolic abnormalities related to the adrenocortical steroids or to impaired carbohydrate metabolism.

Although how obesity contributes to the pathogenesis of arterial hypertension is not entirely clear, from a practical aspect the important point is that the control of obesity helps the control of hypertension.

Epidemiological findings

Epidemiological studies up to 1967 and the more recent ones have been reviewed, respectively, by Chiang et al. (2) and Staessen et al. (3). They provide strong evidence that in populations with "western" life-styles the body weight and blood pressure are positively correlated. This applies also to some African and Asian, and certain traditional ("primitive") societies. The higher prevalence of hypertension in the lower socioeconomic strata of European and North American populations may be explained in part by a three times higher prevalence of obesity in the lower income groups.

Blood pressure and body weight are also correlated in children and adolescents; both variables tend to track, i.e., children who are found to have higher relative body weights and blood pressures than their average peers will continue on the high side even years after the first measurement (4). Age-related increase in blood pressure in the young, like that in middle age, is usually proportional to the gain in body weight. Childhood obesity predisposes to hypertension in adulthood. Therefore, hypertensive children and adolescents may be the ideal target population for the introduction of weight control programmes (5).

These epidemiological associations imply that obesity control has a definite potential for the prevention of hypertension. Thus, in a U.S. community study it has been calculated that, in theory, weight

¹ This text has been approved by the following 27 member organizations of the World Hypertension League: National Heart Foundation of Australia; Belgian Hypertension Committee; British Hypertension Society; Canadian Coalition for Hypertension Control; Danish Society of Hypertension; Comité National de Lutte contre l'Hypertension Artérielle, France; German Hypertension League; Hellenic League against Hypertension; Hypertension Committee of the Hungarian Society of Cardiology; All India Heart Foundation; International Green Cross; Israel Hypertension Society; Italian League against Hypertension; Japan Heart Foundation; Ligue Nationale Marocaine de Lutte contre les Maladies Cardio-Vasculaires; The Netherlands Heart Foundation; Norwegian Society of Hypertension; Polish Society of Hypertension; Portuguese League against Hypertension; Fondation Sénégalaise de Lutte contre l'Hypertension Artérielle; Southern African Hypertension Society; Spanish League against Hypertension; Swedish League against Hypertension; Swiss Association against High Blood Pressure; Liga Uruguaya contra la Hipertension; National High Blood Pressure Education Program, USA; and Zaire League against Hypertension. Requests for reprints or for permission to reproduce this article, or part of it, should be sent to the World Hypertension League, 20 avenue du Bouchet, 1209 Geneva, Switzerland. A summary in French appears on pages 251-252.

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Table 1: Synopsis of 16 studies on weight reduction in hypertensive patients

Study	No. of patients	Duration (months)	Change in body weight (kg)	Change in blood pressure (mmHg)		Result	Drugs	Comments
				Systolic	Diastolic			
<i>Evans County, 1973 (15)*</i>								
Experimental	64	12	-8.2	-18	-13	+	Yes	
Control	63		-1.8	-12	-8			
<i>Glasgow, 1978 (16)</i>								
General advice	20	12	-2.2	-11	-5	+	Yes	Less drugs
Diet information sheet	14		-2.6	-6	-3			
Dietitian	15		-5.1	-12	-7			
<i>Tel Aviv, 1978 (17)</i>								
Weight reduction	24	12-18	-7	-21	-12	+	Yes	No change in drug intake, results independent of salt restriction.
Weight reduction + drugs	57		8	-37	-19			
Drugs only	26							
<i>Los Angeles, 1981 (18)</i>								
Low calorie, low sodium	15	3	-20.2	-19	-17	+	No	Change in mean blood pressure
Low calorie, high sodium	10		-20.2					
<i>UCLA, 1984 (19)</i>								
Low calorie, low sodium	18	3	-6.1	-9		+	No	Results independent of salt restriction.
Low calorie, high sodium	12		-4.7	-9	-5	+	No	
<i>Minnesota, 1983 (20)</i>								
Group I	28	2½	-5.2	-5	-2	+	No	Results independent of salt restriction.
Group II	59		-1.9	-3	-2			
<i>Post HDFP I, 1985 (21)</i>								
Sodium restriction	101	13	-4.0	-5	-2	+	Yes	Significantly more in the experimental group remained without medication.
Weight reduction	87		-7.0					
Control	89		-0.5	-3	-2			

The reference to the study concerned is given in parentheses.

control has a potential of preventing 28% of future cases of essential hypertension in Blacks and 48% in Whites. Of even greater importance is the calculation that by weight control in those who were clearly overweight (12–14% of the population), up to 28% of eventual cases of hypertension *could* be prevented (6). Obesity control thus seems to be a viable approach for the *primary prevention* of many cases of arterial hypertension.

Obviously, not all hypertensives are obese, and not all obese persons have hypertension. Nevertheless, over a period of 10–15 years at least 60% of overweight individuals *will* become hypertensive.

Weight reduction of hypertensive patients

The first observation of blood pressure reduction as a result of weight reduction was published 70 years ago and this has been confirmed subsequently in numerous reports. Out of 16 controlled studies on lowering blood pressure through weight reduction, which were selected for their appropriate methodology and reviewed by the World Hypertension League, 15 were successful to varying degrees (Table 1). However, there is no universal agreement on this issue because of design flaws, one criticism being that the sodium intake was reduced in parallel with the calorie intake; but studies with a reduced calorie intake and constant sodium intake have also shown a decrease in blood pressure, which suggests that the decrease in blood pressure was not caused solely by the diminished sodium intake.

Nevertheless, the evidence in favour of the benefits of reducing overweight is by far outweighing the doubts. Weight control is considered the mainstay of non-pharmacological treatment of hypertension in obese patients. This is supported also by the report of the U.S. Joint National Committee (7). Weight reduction should accompany drug treatment in all obese patients with hypertension, and should be given particular attention in the management of patients with mild blood pressure elevations. If the diastolic blood pressure does not exceed 95 mmHg (12.7 kPa), according to the guidelines of the World Health Organization and the International Society of Hypertension (8), behavioural treatment (i.e., weight control, associated with moderate salt and alcohol intakes and smoking cessation) should be applied. It is understood that, where applicable, emphatic anti-smoking intervention is needed.

Added benefits of weight loss. The impact of weight reduction on the obese hypertensive patient may still not be fully recognized. Experience from the Hypertension Detection and Follow-Up Program in the USA (9) has shown that increases in serum uric acid, glucose, and cholesterol, which are associated with

most diuretic treatment, may be prevented or reduced by concurrent weight loss; in fact, the extent of the change in weight is related to the changes in the serum levels of the above-mentioned substances. In addition, as both hypertension and obesity increase the workload of the heart, their simultaneous occurrence leads to a particularly heavy burden on the left ventricle. Obese hypertensive patients are at high risk of congestive heart failure, coronary heart disease, and sudden death. Even a moderate weight loss may reverse the cardiac hypertrophy and result in other benefits.

Assessment of obesity

The thickness of subcutaneous fat at various points of the body can be estimated using mechanical (constant pressure) or ultrasonic calipers. Total body fatness can be assessed by measuring body density, i.e., weight divided by volume; the latter is measured by immersion or by a tracer method (tritium). However, these methods are reserved for the laboratory. In practice, obesity is equated with overweight, which is defined either as the excess over the sex- and age-related ideal body weight, or as a body mass index (BMI), relating weight to height. The most often used BMI is: weight (kg)/height² (m).

All these measurements result in precise figures. The problem lies in the interpretation of the figures. Like many other measurable characteristics, including blood pressure, obesity is a continuous variable with no natural cut-off between "normal" and "excessive". Arbitrarily, body weights more than 15–20% above the population-related "ideal weight" are usually considered as overweight, i.e., a sign of obesity. However, culture-related values, specific to given societies at a given time, are as important as medical considerations, if not more so; these values may change in time and are associated with societal trends such as fashion.

The diagnosis of obesity, especially in borderline situations, is thus largely subjective. In practice, the conclusion whether obesity is present is based on clinical inspection, confirmed by body weight measurement, or by calculation of the body mass index. Persons with BMI values over 25 are considered as overweight, and those over 30 as obese. For convenience, the terms overweight and obesity are here used interchangeably.

Management of patients

Measurement of blood pressure in obese individuals

Cuffs with a longer and wider rubber bag than normal are needed for measuring blood pressure in

Table 2: Recommended dimensions for the bladder in sphygmomanometer cuffs*

Arm circumference at midpoint ^b (cm)	Type of cuff	Bladder width (cm)	Bladder length (cm)
5-7.5	Newborn	3	5
7.5-13	Infant	5	8
13-20	Child	8	13
17-26	Small adult	11	17
24-32	Adult	13	24
32-42	Large adult	17	32
42-50 ^c	Thigh	20	42

* Recommended by the American Heart Association; data from reference 11.

^b Midpoint of arm is defined as half the distance from the acromion to the olecranon.

^c In persons with very large limbs, the indirect blood pressure should be measured in the leg or forearm.

individuals with a large arm girth. At equal intra-arterial pressure, the standard cuff (12 × 25 cm) needs higher pressures to compress the artery in an obese than in a lean person, thus giving a false high reading (10). Cuffs of 17 cm width have been recommended for obese persons by the American Heart Association (Table 2) (11). The inner bag must be wide enough to cover two-thirds of the length of the upper arm and long enough to cover at least two-thirds of the circumference of the upper arm (12). Conical cuffs were reported to be more suitable for obese arms.

Management guidelines

In cases of gross obesity, all definable and treatable causes should be excluded, in particular obesity of endocrine and hypothalamic etiology (e.g., Cushing syndrome). However, these are rare conditions. In most cases obesity is a result of an imbalance between calorie intake and expenditure. If an individual is in a positive energy balance, the initial response will be an increase in the size of the fat cells (lipocytes) with no increase in their number ("hypertrophic" obesity). There seems to be a critical point where further imbalance between energy intake and expenditure leads to the formation of new mature fat cells ("hyperplastic" obesity).

The principal methods of weight reduction are: (i) change of diet; (ii) behaviour modification; and (iii) exercise. Treatment with appetite-depressing drugs is not recommended because of potential side-effects. Psychotherapy may have supportive value; acupuncture and other methods of "alternative medicine" will not be discussed here.

(i) *Diet*. Since in every case of obesity there is always an imbalance between food intake and

energy expenditure, dietary calorie restriction is a pivotal measure in achieving successful weight reduction. Although the optimal diet for weight reduction has not yet been defined, in principle it is one which contains the least number of calories, yet ensures total or near total preservation in the body of nitrogen, minerals, vitamins and other essential factors. The extent of calorie restriction has to be individually determined, and should be based on age, body size, and physical activity. Generally, the calorie intake of a moderately active female on a long-term weight-reducing regimen should average at 1000-1200 kcal (4200-5000 kJ) per day; for a moderately active man, the intake should aim at 1100 kcal (4600 kJ), with an upper limit of 1400 kcal (5800 kJ) per day. The need for individual assessment cannot be overemphasized. Depending on the level of energy expenditure, these values may only rarely be shifted upward.

It is essential that the diet should be balanced, with an appropriate regular daily intake of legumes, vegetables, grains, fish and poultry, fruits, and skim milk products.

A particularly dangerous diet for the hypertensive patient is the so-called "low-carbohydrate diet" with less than 50 g of carbohydrates per day. Such a diet, apart from causing severe ketosis and therefore being totally inappropriate for patients with diabetes mellitus, renal disease and other acidotic states, frequency leads to an increased fat intake which is highly undesirable in hypertensive patients where the multi-risk management of elevated blood pressure and, in particular, an appropriate lipoprotein spectrum are essential to prevent cardiovascular incidents.

It should be added that crash diets (such as the 'Scarsdale diet', 'banana diet', 'macrobiotic diet', etc.) should be avoided. Even though they may lead to temporary weight loss, they will not solve the long-term problem of obesity.

(ii) *Behaviour modification*. It is now generally recognized that a desirable change in the diet and long-term improvement of the balance between calorie intake and expenditure cannot be achieved without an appropriate mental attitude. Thus, changing the obese patient's way of thinking and attitude towards food is the second pillar on which any successful dietary intervention rests.

The methods used for behaviour modification are varied. Most techniques comprise step-wise education of the overweight patient by providing, in the first phase of the intervention, a feedback requesting the patient to record exactly the amount and type of food consumed and the circumstances under which this occurred. The next phase aims at achieving a

regular meal intake at defined times, creating an appropriate atmosphere for the consumption of meals, changing eating and shopping habits, and providing emotional support for sustaining the new regimen.

(iii) *Exercise.* Obese adults usually have a lower level of habitual physical exercise than control groups. Although the benefits of exercise alone in terms of energy expenditure are limited (e.g., it takes 46 minutes of cycling to burn off the caloric gain from one slice of apple pie), increased physical activity is nevertheless considered to be an important adjunct to the treatment of obesity. Exercise leads to improvements in cardiac function and glucose metabolism, and to psychological benefits such as improved self-esteem.

Clinical management. This should be based on the premise that the fall in blood pressure after weight reduction is the consequence of both salt loss and weight reduction, rather than of weight reduction alone. The following essential points should be observed:

- A weight-reducing, low-calorie diet should be vigorously recommended.
- The diet should include restriction of sodium to about 70 milliequivalents per day.
- In severe and moderate hypertension weight reduction alone does not suffice, and drug treatment should be instituted together with diet. In mild hypertension (diastolic blood pressures up to 95 mmHg on repeated occasions), only non-pharmacological treatment is recommended.

Practical aspects of weight control

- *The physician's role.* Without the direct involvement and categorical endorsement by the patient's treating physician, the efficacy of any weight-reducing intervention will be limited and declining. It is therefore essential that physicians who have to treat overweight hypertensive patients are well acquainted with the techniques of weight reduction.
- *Teamwork.* It is difficult for the busy physician to handle alone the complex issues of behavioural and nutritional interventions. Teams consisting of a dietician, nursing assistant, and health educator providing supportive services have been both effective and efficient, when adequately backed up by the treating physician. These allied health personnel may not only have more time for counselling the patient, they can also advise and educate family members, particularly those who prepare the patient's meals.
- *Group therapy.* Since counselling of overweight patients is time-consuming, it is practical to organize

group sessions; in addition, and even more importantly, mutual encouragement among members of the group and other aspects of group dynamics play a major role in the successful maintenance of a treatment programme.

- *Interprofessional collaboration.* Positive experience has been described with interprofessional collaboration. Dentists, optometrists, pharmacists and podiatrists have been involved with success in the National High Blood Pressure Program in the USA. Having ample opportunities for interacting with the patients, these health professionals were able to give additional support to obese hypertensive patients and to encourage them to keep up with the weight control regimen (13).

- *Teaching aids.* Health counselling should be supported by printed material and, if possible, by video tapes, food tasting sessions, special cooking classes, shopping instructions, advice on eating out, etc. If possible, the physician should review the contents of the teaching material with the patient. The teaching aids should be evaluated for scientific accuracy and readability; their comprehension by patients should be assessed and, ideally, also their impact on the behaviour of the target groups. Ineffective, unpalatable or inappropriate teaching aids should be abandoned.

- *Comprehensiveness.* Weight control is only one aspect of what should be a well-balanced, comprehensive programme for the management of hypertensive patients. As these individuals mostly live with their families, there is a need to include the mother or spouse and, if necessary, other family members into the overweight management programme. In a number of populations obesity is not only an individual or family problem, but a societal phenomenon as well. The encouragement of weight control thus has wider, social dimensions, and the prevention of obesity may well contribute to the primary prevention of many cases of hypertension.

Since the aim of the present statement is to provide guidance to medical practitioners in their daily activities, the establishment of community-orientated education programmes will not be discussed here. However, a harmonious social environment should be noted as a crucial factor for successful management of individual obese hypertensive patients.

- *Simplicity and specificity.* Complex dietary advice is usually not understood, and the patients feel intimidated by the difficulties. A simple system of instructions, not longer than a page, should be developed, taking into account the sociocultural environment of the patient, pertinent cooking and

eating habits, availability of various food items, and the cost. These factors vary widely geographically and sociopolitically, even between social classes within one population.

● *Improving adherence.* Patients often have difficulties in following medical advice because of misconceptions; it is useful to identify these and to clarify the patients' problems (14). Many patients are discouraged because they always seem to regain the weight they lost. Others think that weight-reducing diets are necessarily boring. Many patients feel a lack of will-power to lose weight. They find it difficult to resist snacks and may feel that their diet is too restrictive.

All these patients need reassurance concerning their specific problem. The physician should identify the problem and make it clear that weight control needs effort in the long run, that dieting needs to be combined with physical exercise, that calorie intake should be reduced to 1000–1200 kcal per day but not lower, and that low-calorie diets can be made tasty and enjoyable (15). The patient's will-power may be reinforced by participation in group sessions.*

● *Recidivism.* Even with full success in reducing body weight, some individuals after a time regain or even exceed the original weight. The rate of recidivism is extremely high; special attention must therefore be paid to regular follow-up and continuing reinforcement of motivation for long-term adherence of patients who have successfully reduced their weight.

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Résumé

Prise en charge de l'hypertension et lutte contre la surcharge pondérale

Le présent article, préparé par la Ligue mondiale contre l'hypertension, fait le point des connaissances actuelles sur ce sujet complexe et fournit des indications à l'intention des médecins.

L'obésité est un facteur important souvent associé à la pathogenèse de l'hypertension artérielle essentielle, le poids et la tension artérielle montrant une corrélation positive aussi bien chez l'enfant que chez l'adulte. Chez le jeune comme chez la personne d'âge mûr l'augmentation de la tension artérielle en fonction de l'âge est en général proportionnelle au gain de poids. L'obésité au cours de l'enfance prédispose à une hypertension à l'âge adulte. Par conséquent, les enfants et les adolescents hypertendus représentent probablement la population cible idéale pour des programmes de lutte contre la surcharge pondérale. La lutte contre l'obésité constitue une approche intéressante de la prévention primaire dans de nombreux cas d'hypertension artérielle.

La lutte contre la surcharge pondérale est l'élément de base du traitement non pharmacologique des hypertendus obèses. Elle doit aller de pair avec le traitement médicamenteux en cas d'hypertension sévère et modérée, et être associée à d'autres méthodes non pharmacologiques ainsi qu'à un programme d'éducation permanente en cas d'hypertension légère. Pour un hypertendu obèse, la perte de poids n'entraîne pas seulement un abaissement de la tension artérielle; elle comporte plusieurs autres avantages métaboliques et hémodynamiques.

En pratique, la meilleure façon d'évaluer l'obésité consiste à faire un examen clinique et à calculer l'indice de masse corporelle (IMC). La mesure de la tension artérielle chez l'obèse nécessite l'utilisation d'un brassard plus large que dans les autres cas. On obtient une diminution de poids par un régime alimentaire associé à une modification du comportement et à un exercice physique. Il est indispensable d'équilibrer le régime alimentaire, en prévoyant un apport quotidien régulier et suffisant de légumes, de céréales, de poisson et de volaille, de fruits et de produits laitiers écrémés. Il faut éviter les régimes irréguliers ou draconiens. On estime que l'augmentation de l'activité physique est un appoint important dans le traitement de l'obésité. L'exercice physique améliore la fonction cardiaque et le métabolisme du glucose et il est bénéfique sur le plan psychologique en donnant par exemple au sujet une meilleure opinion de lui-même.

Les aspects pratiques de la prise en charge clinique sont les suivants: mise en route et suivi du programme de lutte contre la surcharge pondérale par le médecin traitant, travail d'équipe, thérapie de groupe, collaboration interprofessionnelle et fourniture de matériel éducatif. Pour être complet, un programme de lutte contre la surcharge pondérale doit impliquer les membres de

* A 36-page guide for physicians on improving adherence among hypertensive patients (14) can be obtained, on request, from the National High Blood Pressure Education Program, 120/80 National Institutes of Health, Bethesda, MD 20882, USA, or from the World Hypertension League, 20 Avenue du Bouchet, 1209 Geneva, Switzerland.

la famille du malade et, au-delà, la communauté pour la création d'un environnement social harmonieux. Il doit être à la fois simple et spécifique et l'on accordera une attention particulière à l'adhésion des malades au programme, afin d'éviter les rechutes.

References

1. **World Hypertension League.** Self-measurement of blood pressure. *Bulletin of the World Health Organization*, **66**: 155–159 (1988) and *Journal of hypertension*, **6**: 257–261 (1988).
2. **Chiang, B.N. et al.** Overweight and hypertension: a review. *Circulation*, **39**: 403–421 (1969).
3. **Staessen, J. et al.** Blood pressure, calorie intake and obesity. In: Bulpitt, C.J., ed. *Handbook of hypertension*, Vol. 6. *Epidemiology of hypertension*. Amsterdam, Elsevier, 1985.
4. **Török, E. et al.** Associated factors and tracking of blood pressure in adolescents. In: Strasser, T. & Ganten, D., ed. *Mild hypertension—from drug trials to practice*. New York, Raven Press, 1987.
5. Report of the Second Task Force on Blood Pressure Control in Children. *Pediatrics*, **79**: 1–15 (1987).
6. **Tyroler, H.A. et al.** Weight and hypertension: Evans County studies of blacks and whites. In: Paul, O., ed. *Epidemiology and control of hypertension*. Chicago, Year Book Medical, 1975.
7. The 1988 Report of the Joint National Committee on Detection, Evaluation, and Treatment of High Blood Pressure. *Archives of internal medicine*, **148**: 1023–1038 (1988).
8. 1986 Guidelines for the treatment of mild hypertension: Memorandum from a WHO/ISH Meeting. *Bulletin of the World Health Organization*, **64**: 31–35 (1986).
9. **Heyden, S. et al.** The relationship of weight change to changes in blood pressure, serum uric acid, cholesterol and glucose in the treatment of hypertension. *Journal of chronic diseases*, **38**: 281–288 (1985).
10. **Maxwell, M.H. et al.** Error in blood-pressure measurement due to incorrect cuff size in obese patients. *Lancet*, **2**: 33–36 (1982).
11. **Kirkendall, W.M. et al.** Recommendations for human blood pressure determination by sphygmomanometers. *Circulation*, **62**: 1146A–1155A (1980).
12. WHO Technical Report Series, No. 622, 1978 (*Arterial hypertension: Report of a WHO Expert Committee*).
13. **Coordinating Committee of the National High Blood Pressure Education Program.** Collaboration in high blood pressure control among professionals and with the patient. *Annals of internal medicine*, **101**: 393–395 (1984).
14. **Working Group on Health Education and High Blood Pressure Control.** *The physicians' guide: improving adherence among hypertensive patients*. U.S. Department of Health and Human Services, National Institutes of Health, Bethesda, MD, 1987.
15. **Heyden, S. et al.** Diet treatment of obese hypertensives. *Clinical science and molecular medicine*, **45**: 209–212 (1973).
16. **Ramsay, L.E. et al.** Weight reduction in a blood pressure clinic. *British medical journal*, **2**: 244–245 (1978).
17. **Reisin, E. et al.** Effect of weight loss without salt restriction on the reduction of blood pressure in overweight hypertensive patients. *New England journal of medicine*, **298**: 1–5 (1978).
18. **Tuck, M.L. et al.** The effect of weight reduction on blood pressure, plasma renin activity, and plasma aldosterone levels in obese patients. *New England journal of medicine*, **304**: 930–933 (1981).
19. **Maxwell, M.H. et al.** Blood pressure changes in obese hypertensive subjects during rapid weight loss—Comparison of restricted v. unchanged salt intake. *Archives of internal medicine*, **144**: 1581–1584 (1984).
20. **Gillum, R.F. et al.** Nonpharmacologic therapy of hypertension: the independent effects of weight reduction and sodium restriction in overweight borderline hypertensive patients. *American heart journal*, **105**: 128–133 (1983).
21. **Langford, H.G. et al.** Dietary therapy slows the return of hypertension after stopping prolonged medication. *Journal of the American Medical Association*, **253**: 657–664 (1985).
22. **MacMahon, S.W. et al.** Comparison of weight reduction with metoprolol in treatment of hypertension in young overweight patients. *Lancet*, **1**: 1233 (1985).
23. **Haynes, R.B. et al.** Failure of weight reduction to reduce mildly elevated blood pressure: a randomized trial. *Journal of hypertension*, **2**: 535–539 (1984).
24. **Fagerberg, B. et al.** Blood pressure control during weight reduction in obese hypertensive men: separate effects of sodium and energy restriction. *British medical journal*, **288**: 11–14 (1984).
25. **Zidek, W. et al.** Weight reduction and salt restriction in hypertension: effects on blood pressure and intracellular electrolytes. *Klinische Wochenschrift*, **64**: 1183–1185 (1986).
26. **Stamler, R. et al.** Trial on control of hypertension by nutritional means: three-year results. *Journal of hypertension*, **2** (suppl. 3): 167–170 (1984).
27. **Imai, Y. et al.** Effect of weight loss on blood pressure and drug consumption in normal weight patients. *Hypertension*, **8**: 223–228 (1986).
28. **Cohen, N. & Flamenbaum, W.** Obesity and hypertension: demonstration of a "floor effect". *American journal of medicine*, **80**: 177–181 (1986).
29. **MacMahon, S.W. et al.** The effect of weight reduction on left ventricular mass—A randomized controlled trial in young, overweight hypertensive patients. *New England journal of medicine*, **314**: 334–339 (1986).